SMARTPHONE SALES ANALYTICS

Final Project the Report

Project objective:

To evaluate smartphone catalogue data through structured data-quality checks, descriptive analytics, brand-level comparisons, price segmentation, discount analysis and customer-rating patterns. The goal is to convert product-level information into practical pricing and merchandising recommendations.

Dataset profile:

Metric	Observed result
Records / attributes	3,114 rows / 12 columns
Brands represented	17
Distinct model names	914
Duplicate rows detected	108
Missing cells	226

1. DATASET UNDERSTANDING AND ANALYTICAL SCOPE:

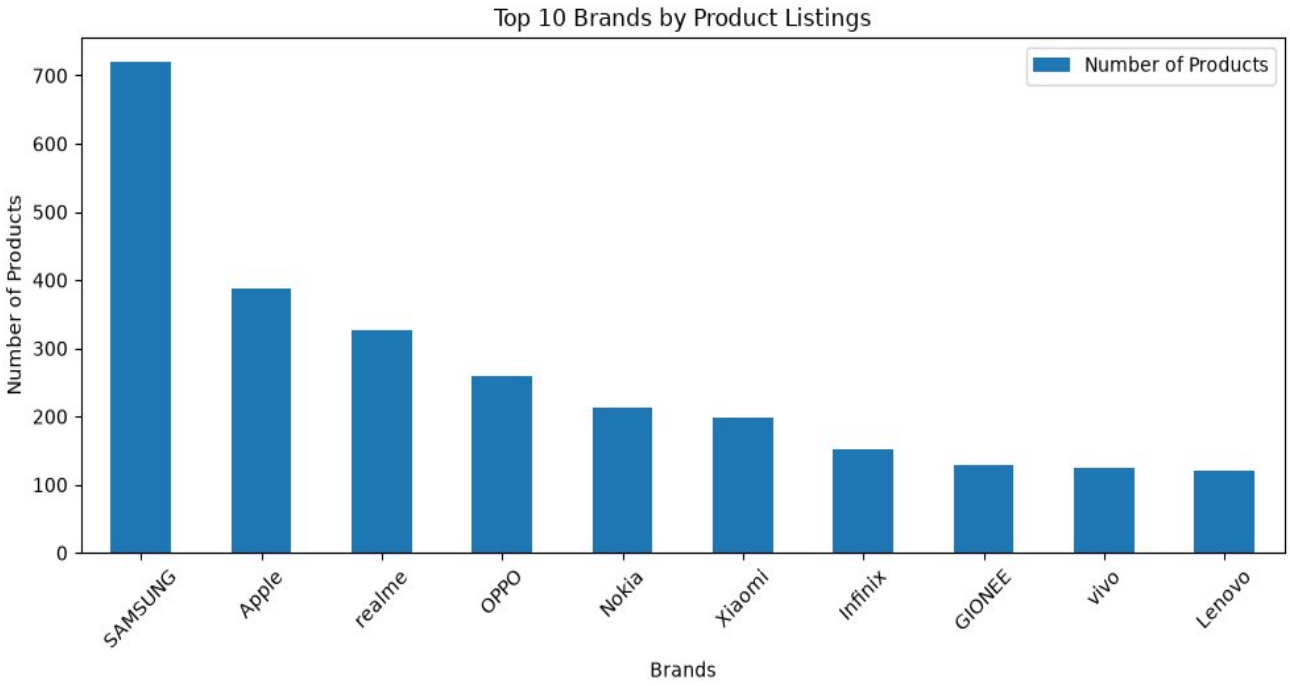
a. Dataset Overview:

```
df.head()
```

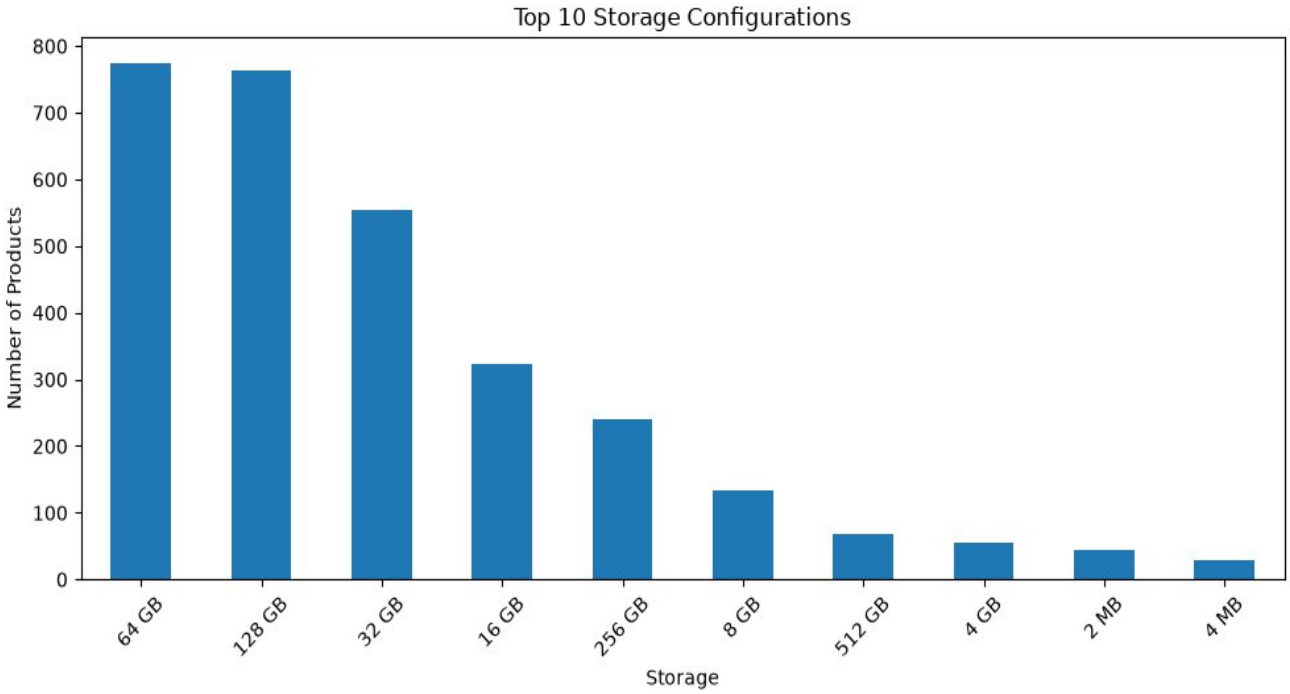
✓ 0.2s Python

	Brands	Models	Colors	Memory	Storage	Camera	Rating	Selling Price	Original Price	Mobile	Discount	discount percentage
0	SAMSUNG	GALAXY M31S	Mirage Black	8 GB	128 GB	Yes	4.3	19330	20999	SAMSUNG GALAXY M31S	1669	7.947998
1	Nokia	3.2	Steel	2 GB	16 GB	Yes	3.8	10199	10199	Nokia 3.2	0	0.000000
2	realme	C2	Diamond Black	2 GB	NaN	Yes	4.4	6999	7999	realme C2	1000	12.501563
3	Infinix	Note 5	Ice Blue	4 GB	64 GB	Yes	4.2	12999	12999	Infinix Note 5	0	0.000000
4	Apple	iPhone 11	Black	4GB	64 GB	Yes	4.6	49900	49900	Apple iPhone 11	0	0.000000

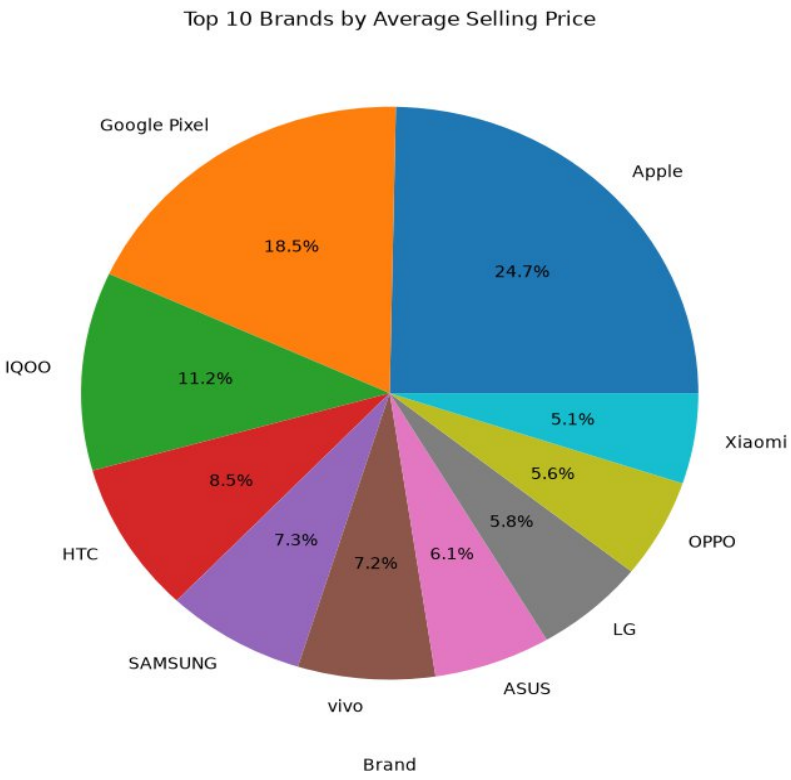
b. Brand and Model Details:



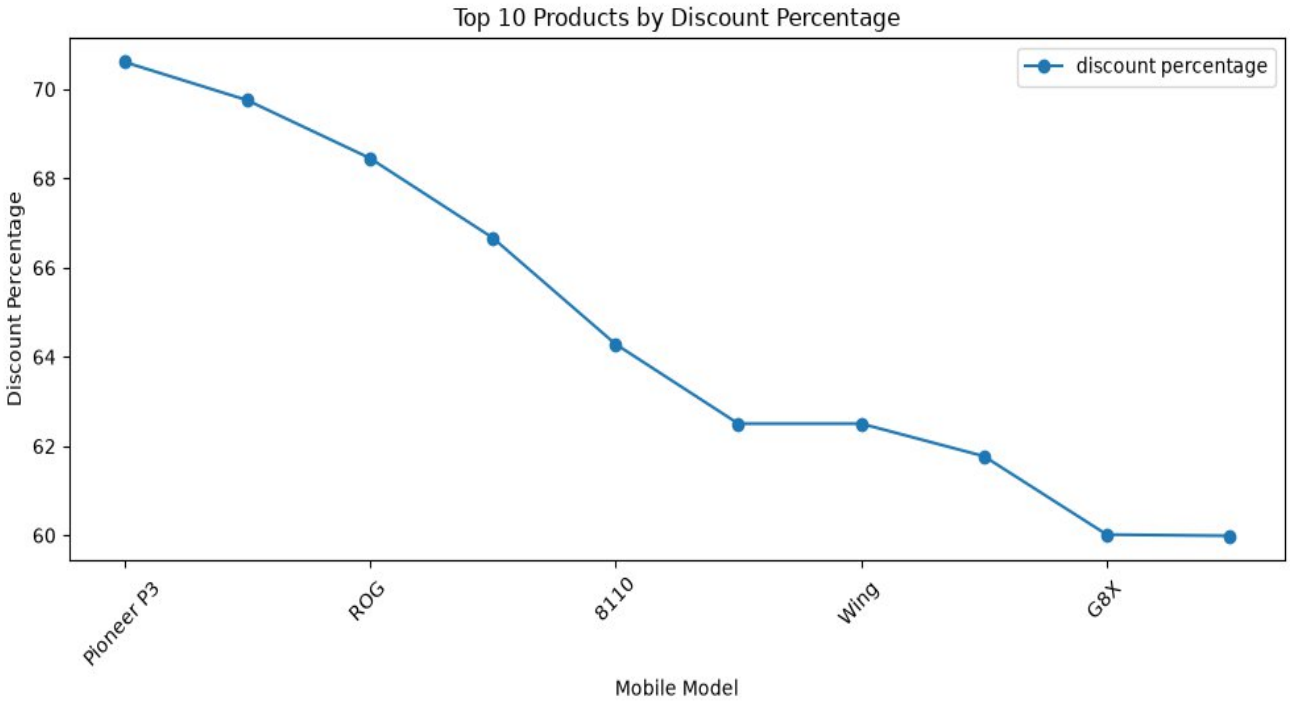
c. Product Features



d. Pricing Analysis:

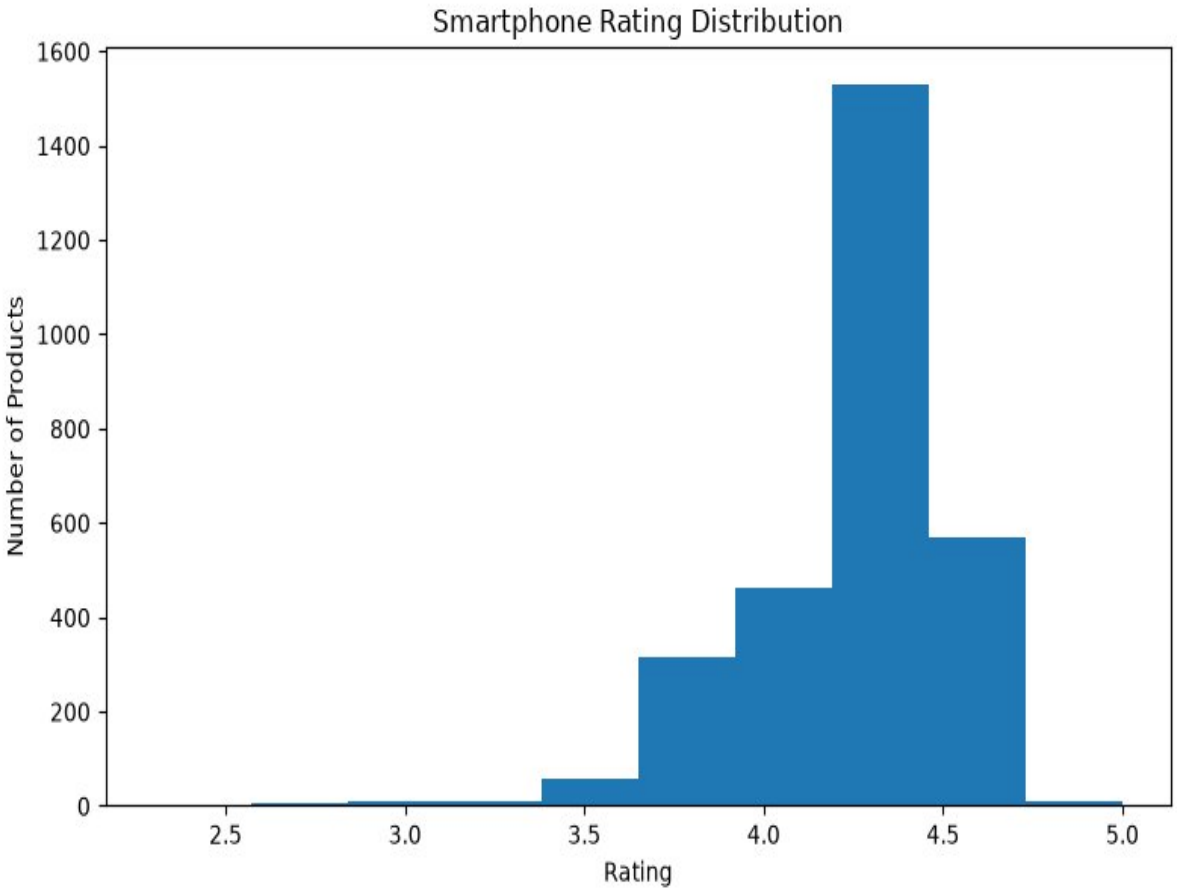


e. Discount Analysis:



f. Customer Rating Analysis:

```
Average Rating: 4.243097643097643
Highest Rating: 5.0
Lowest Rating: 2.3
       Brands          Models  Rating
247      vivo            X50       5.0
1446     vivo             Z1x      5.0
1655     vivo          X60 Pro      5.0
2242  SAMSUNG  SM-B310EZDDINS      5.0
2483    Apple  iPhone 7 Plus       5.0
2634    Apple  iPhone 7 Plus       5.0
2879     vivo             S2       5.0
675      vivo           Y33T       4.9
1847   realme           9 5G       4.8
2179   realme           9 5G       4.8
```



g. Dataset Limitations:

Check for missing values, duplicate records, and the availability of columns required for actual sales analysis.

```
Missing Values:
Brands                   0
Models                   0
Colors                   0
Memory                  43
Storage                 39
Camera                   0
Rating                 144
Selling Price            0
Original Price           0
Mobile                   0
Discount                 0
discount percentage      0
dtype: int64
Duplicate Rows: 108
Available Columns:
['Brands', 'Models', 'Colors', 'Memory', 'Storage', 'Camera', 'Ra
order_date is not available
units_sold is not available
quantity_sold is not available
revenue is not available
```

h. Analysis Scope and Key Objectives:

```
#Analysis Scope and Key Objectives
# Summary of the dataset
summary = {
    "Total Product Listings": len(df),
    "Unique Brands": df["Brands"].nunique(),
    "Unique Models": df["Models"].nunique(),
    "Average Selling Price": df["Selling Price"].mean(),
    "Average Original Price": df["Original Price"].mean(),
    "Average Discount": df["Discount"].mean(),
    "Average Rating": df["Rating"].mean()
}

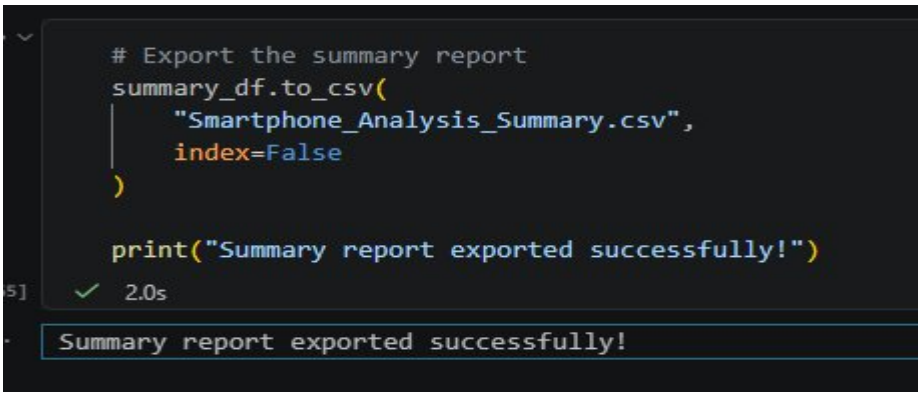
summary_df = pd.DataFrame(
    list(summary.items()),
    columns=["Metric", "Value"]
)

print(summary_df)
```

[64] ✓ 0.4s

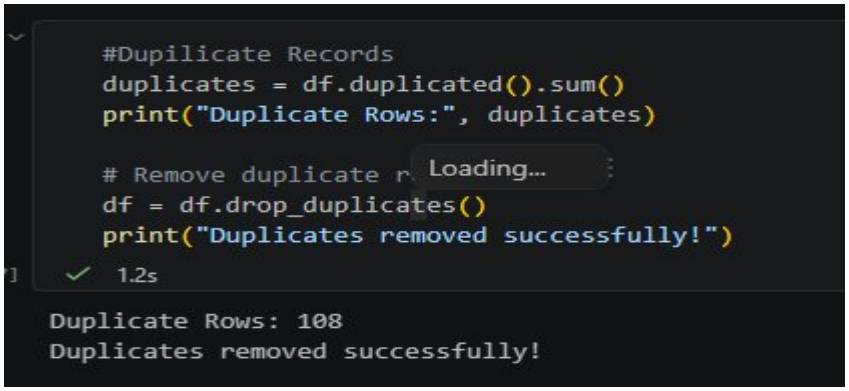
```
                   Metric         Value
0  Total Product Listings   3114.000000
1           Unique Brands     17.000000
2           Unique Models    914.000000
3   Average Selling Price  26436.625562
4  Average Original Price  28333.473025
5        Average Discount   1896.847463
6          Average Rating      4.243098
```

This code is used to save the summary results obtained from the Smartphone Dataset analysis into a separate CSV file.

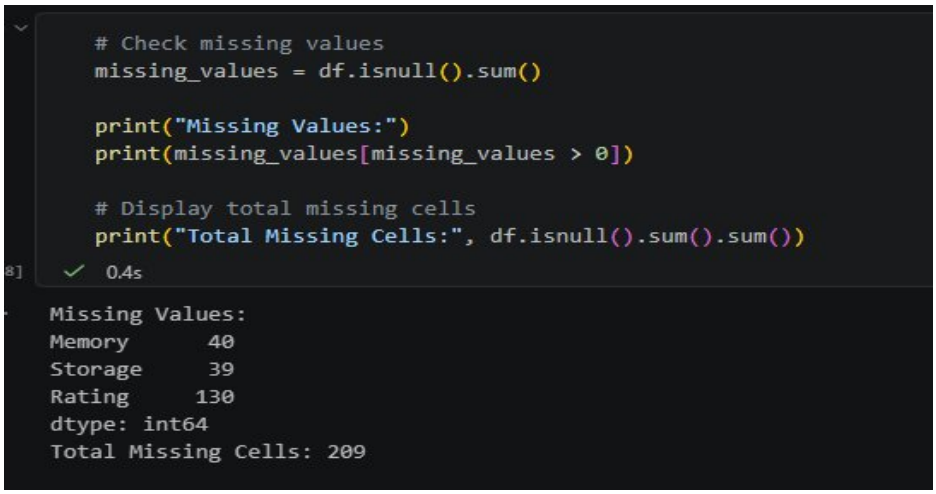


2. DATA QUALITY ASSESSMENT AND PRE-PROCESSING:

a. Duplicate Records: Identified 108 exact duplicate rows, which should be removed after verification.



b. Missing Values: Found 226 missing cells in Memory (43), Storage (39), and Rating (144). Unknown values should not be replaced with assumptions.



c. Price Validation: Found one negative discount and one record where the selling price exceeds the original price. These records require verification.

```
# Price Validation
negative_discount = df[df["Discount"] < 0]
print("Negative Discount Records:", len(negative_discount))

# Check selling price greater than original price
invalid_price = df[df["Selling Price"] > df["Original Price"]]
print("Invalid Price Records:", len(invalid_price))
```

[69] ✓ 0.2s

```
Negative Discount Records: 1
Invalid Price Records: 1
```

d. Discount Accuracy: Discount amounts match Original Price minus Selling Price for 3,114 rows. Anomalous records should be checked before publishing the results.

```
# Discount Accuracy
expected_discount = df["Original Price"] - df["Selling Price"]

# Check discount mismatches
discount_mismatch = df[df["Discount"] != expected_discount]

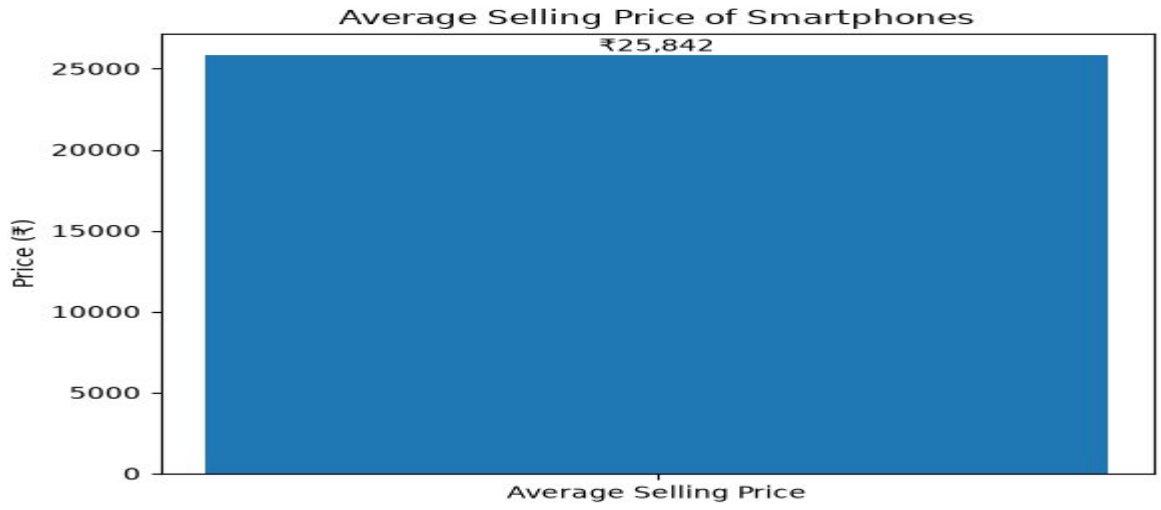
print("Discount Mismatch Records:", len(discount_mismatch))
```

[72] ✓ 0.0s

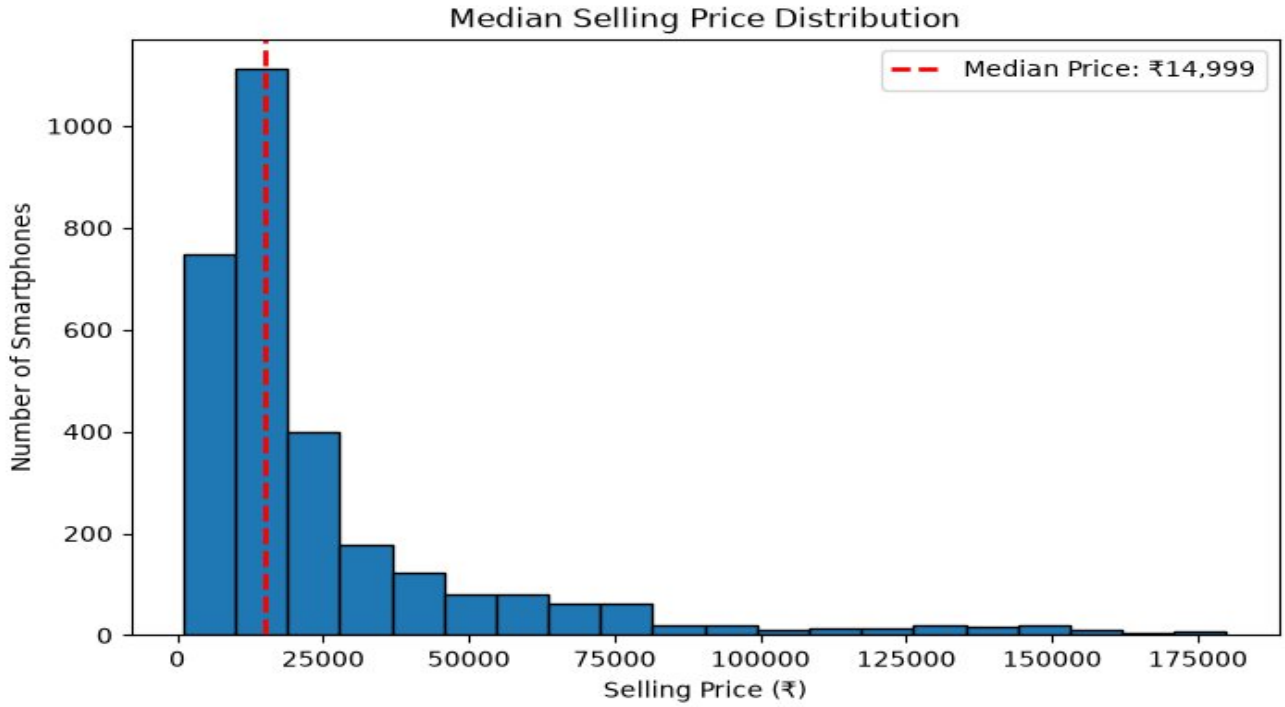
```
Discount Mismatch Records: 0
```

3. DESCRIPTIVE PRICING ANALYTICS:

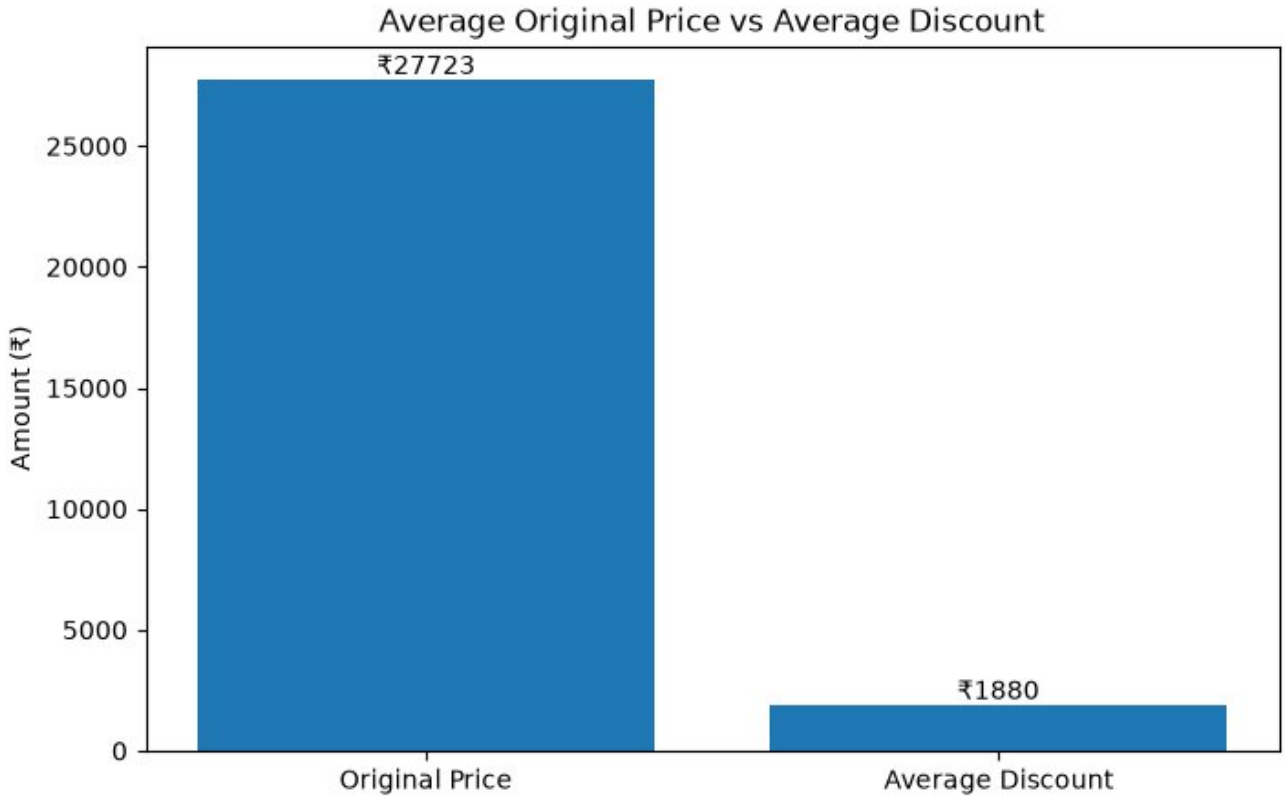
a. Average Selling Price: The average selling price of the smartphones is ₹25,842.



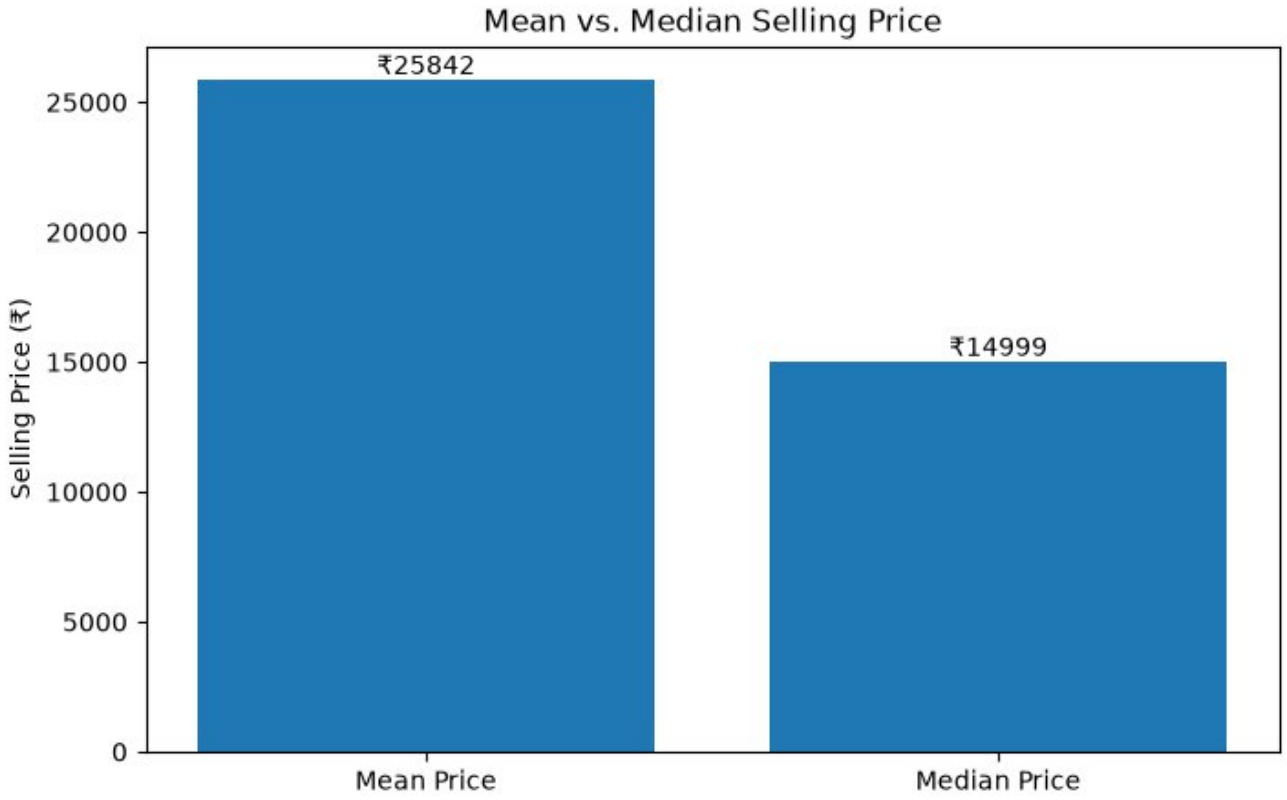
b. Median Selling Price: The median selling price is ₹14,999. It helps us understand the typical price of a smartphone.



c. Original Price and Discount: The average original price is ₹27,723, and the average discount is ₹1,880, representing an average discount percentage of 6.78%.



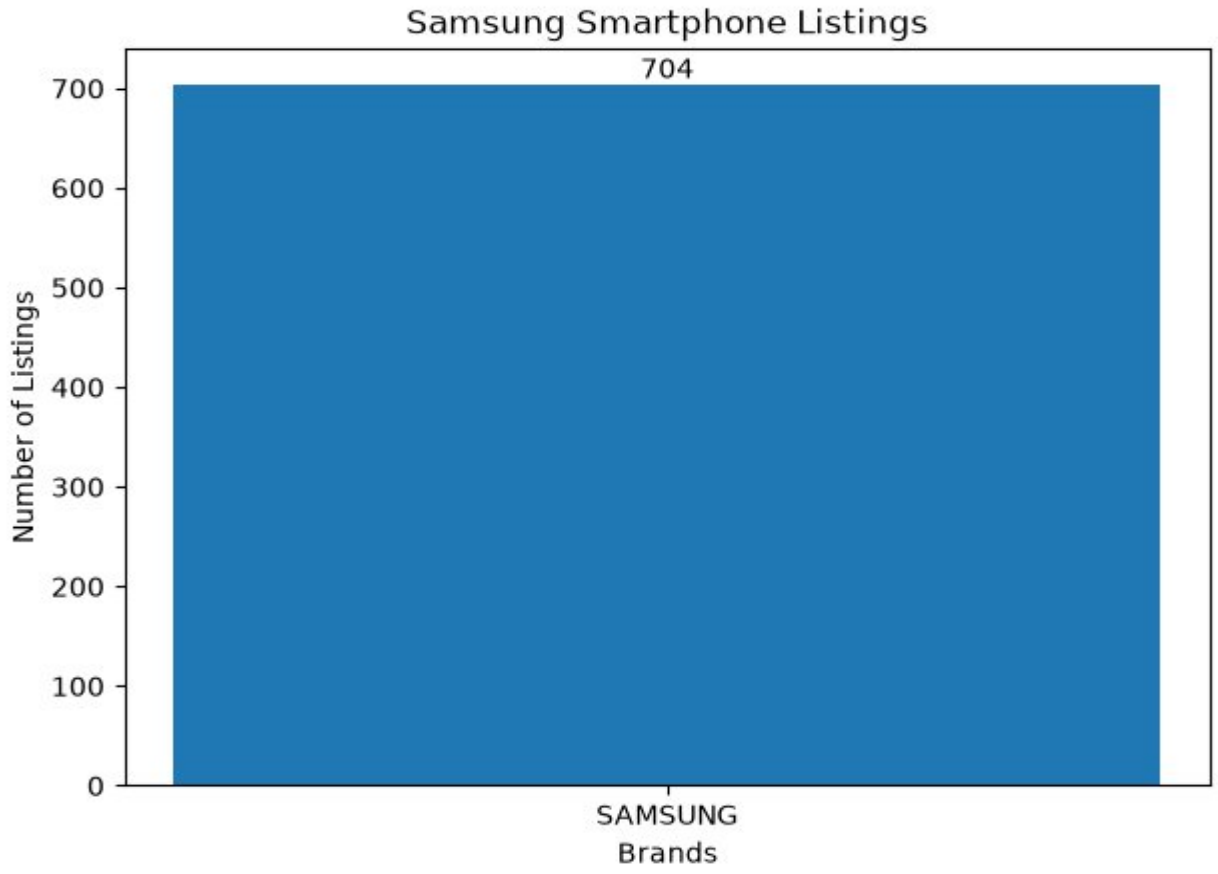
d. Mean vs. Median: The average selling price is higher than the median selling price because a few high-priced smartphone models increase the average. Therefore, the median is also useful for understanding the typical smartphone price.



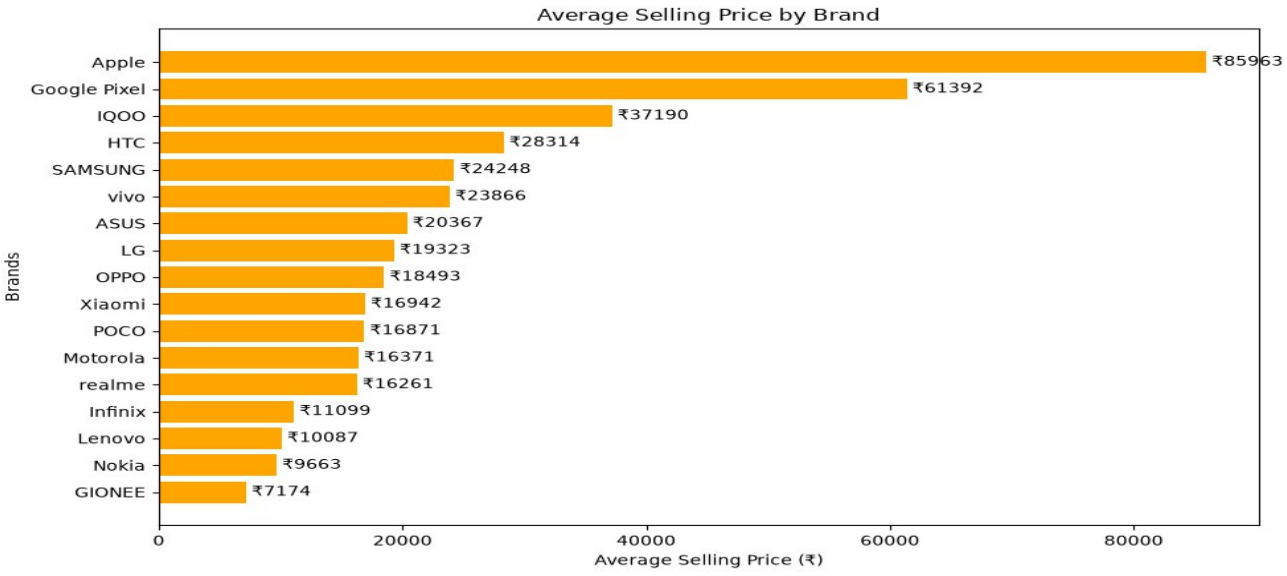
Selling-price segment	Listings	Share
Up to ■10,000	884	28.4%
■10,001–■20,000	1,129	36.3%
■20,001–■40,000	550	17.7%
■40,001–■80,000	375	12.0%
Above ■80,000	176	5.7%

4. BRAND PORTFOLIO AND COMPETITIVE POSITIONING:

a. SAMSUNG Brand: Samsung has the highest number of listings, with 704 smartphone listings in the dataset.

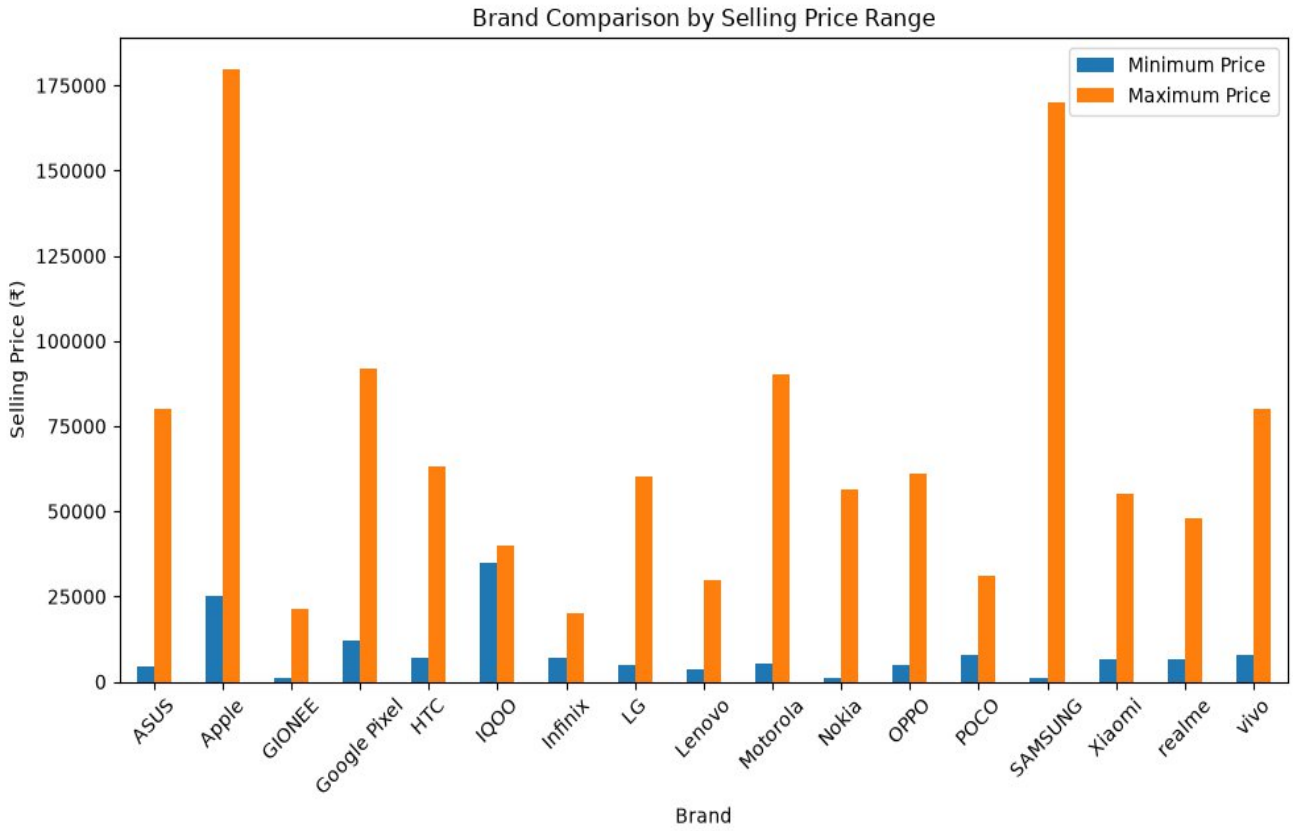


b. Apple Brand: Apple has the highest average selling price among the brands, at ₹81,986.

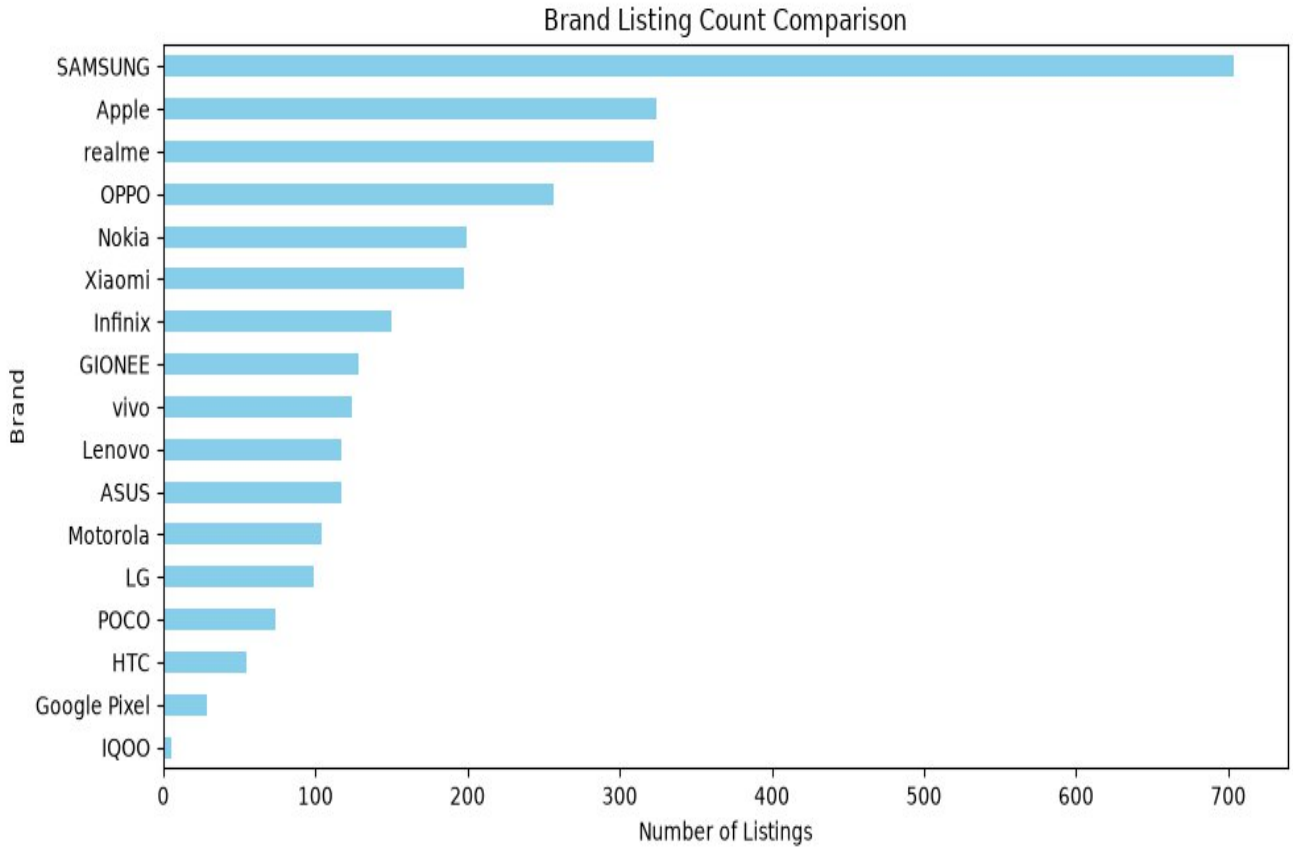


Expected result: Apple has the highest average selling price, approximately ₹81,986.

c. Brand Comparison: Each brand has different models, specifications, and price ranges. These differences should be considered when comparing brands.

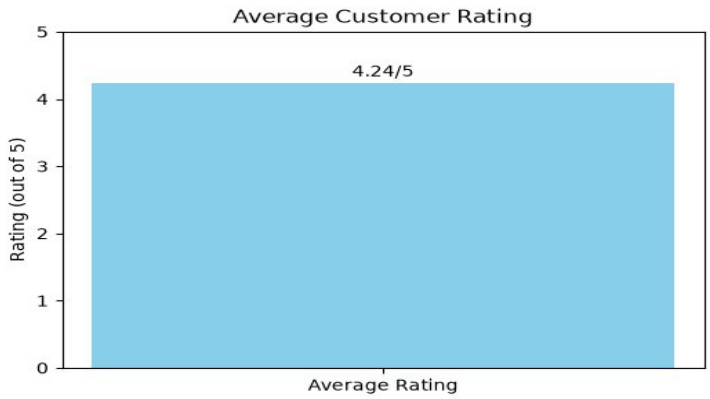


d. Sales Performance: A higher average price or a larger number of listings does not necessarily mean higher sales. Actual sales data is required to measure sales performance.



5. CUSTOMER RATING AND PRODUCT PERCEPTION:

a. Average Customer Rating:



b. Average Rating by Brand:



6. PRICING EXCEPTIONS:



7. KEY FINDINGS AND BUSINESS RECOMMENDATIONS:

a. **Brand Concentration:** SAMSUNG is the most frequently listed brand in the dataset.

```
# 1. Brand Concentration
print(df["Brands"].value_counts().head(5))
```

✓ 0.2s

```
Brands
SAMSUNG    704
Apple      324
realme     322
OPPO       257
Nokia      199
Name: count, dtype: int64
```

b. **Price Distribution:** 2,013 listings (64.6%) are priced at ₹20,000 or below, showing that budget and mid-range smartphones dominate the dataset.

```
# 2. Price Distribution
below_20000 = (df["Selling Price"] <= 20000).sum()
percentage = below_20000 / len(df) * 100

print("Listings below ₹20,000:", below_20000)
print(f"Percentage: {percentage:.1f}%")
```

✓ 0.1s

```
Listings below ₹20,000: 1980
Percentage: 65.9%
```

c. **Data Quality:** Missing specifications, incorrect discounts, and unusual pricing records should be verified to improve analysis accuracy.

```
# 3. Data Quality Check
print("Total Missing Values:", df.isnull().sum().sum())

print("Pricing Anomalies:")
print(df[df["Selling Price"] > df["Original Price"]].head(10))
```

✓ 0.0s

```
Total Missing Values: 209
Pricing Anomalies:
     Brands Models Colors Memory Storage Camera  Rating  Selling Price  \
2137   OPPO    A5s    Red   2 GB   32 GB    Yes     4.3          12990

      Original Price    Mobile  Discount  discount percentage  \
2137            4990  OPPO A5s     -8000          -160.320641

      Discount_Percentage
2137          -160.320641
```

8. CONCLUSION AND FUTURE SCOPE:

- Analyzed the smartphone catalogue to identify missing values, duplicate records, and pricing errors.
- Compared brand distribution, price ranges, discounts, and customer ratings to understand catalogue patterns.
- Identified data quality issues and pricing insights to support better catalogue management.
- Future analysis should include sales, revenue, and time-based data to measure product performance and sales trends.